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Affective Vision Computing Lab

From VLM to VLA: Concepts, Architectures, Data & Training Methods

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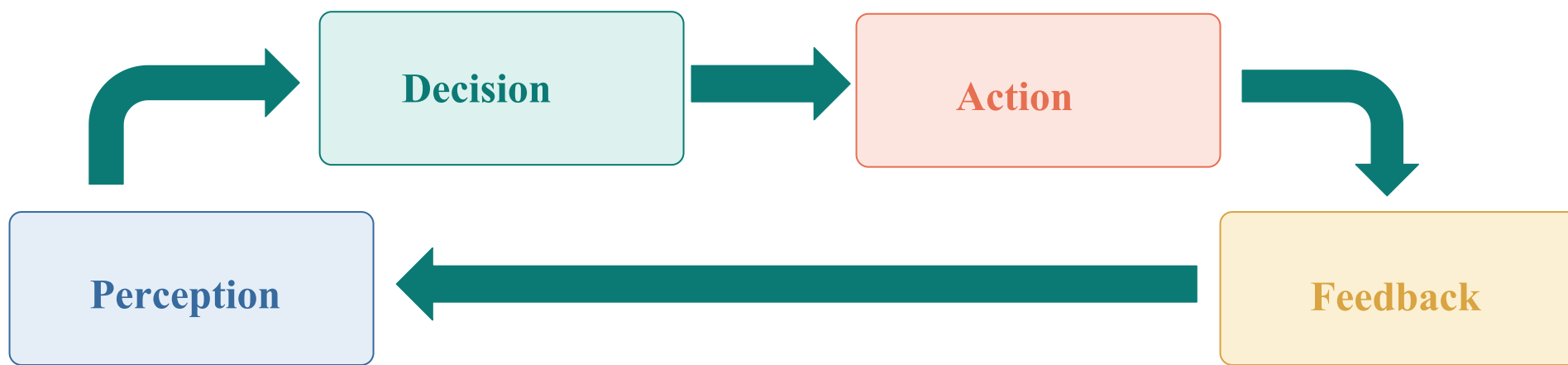
04 **Understand the Training Methods**

01

Define Embodied AI, VLM, VLA

What is Embodied AI?

- ❑ Embodied AI must possess a **physical embodiment**.
- ❑ It actively **interacts with the physical environment**.
- ❑ Distinct from Conversational AI and video-generating Generative AI.
- ❑ Core: Building a complete **Perception-Decision-Action-Feedback** physical loop.



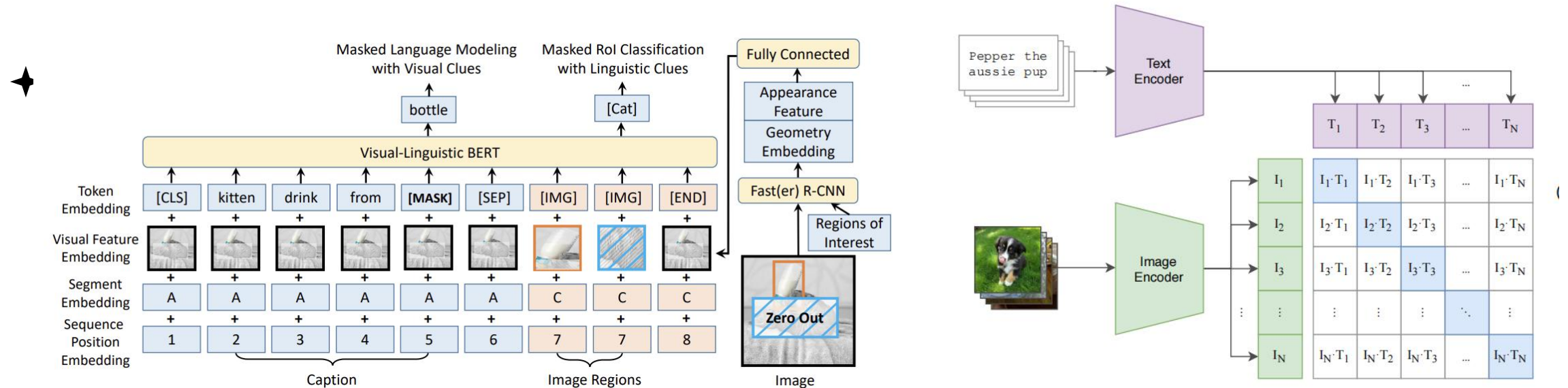


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What is VLM?

- ✦ The fusion of **computer vision** and **natural language processing** models.
- ✦ Tasks: image captioning, visual question answering, visual grounding, etc.
- ✦ Architecture Evolution: CNNs & RNNs → Transformer-based high-capacity models.
- ✦ Key Pretraining:
 - Self-supervised Pretraining (single-stream and multi-stream) e.g., VL-BERT, ViLBERT



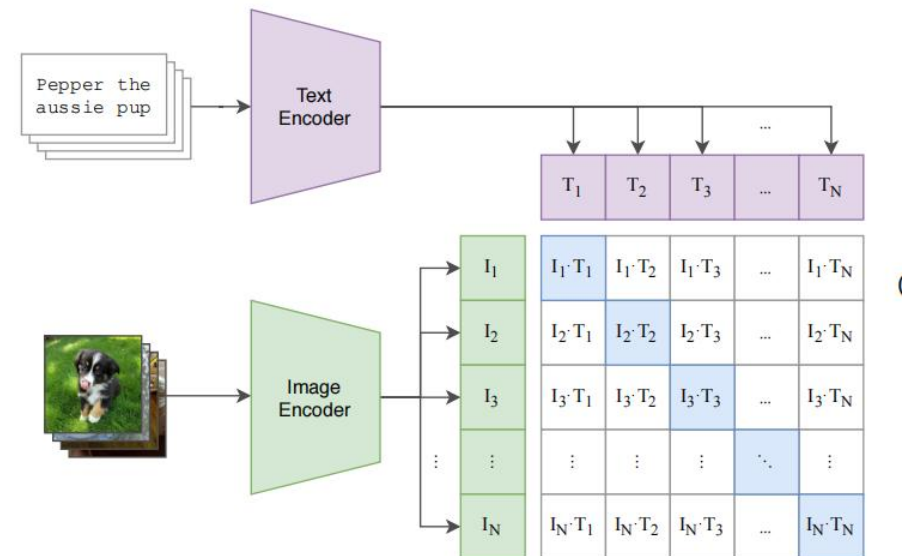


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What is VLM?

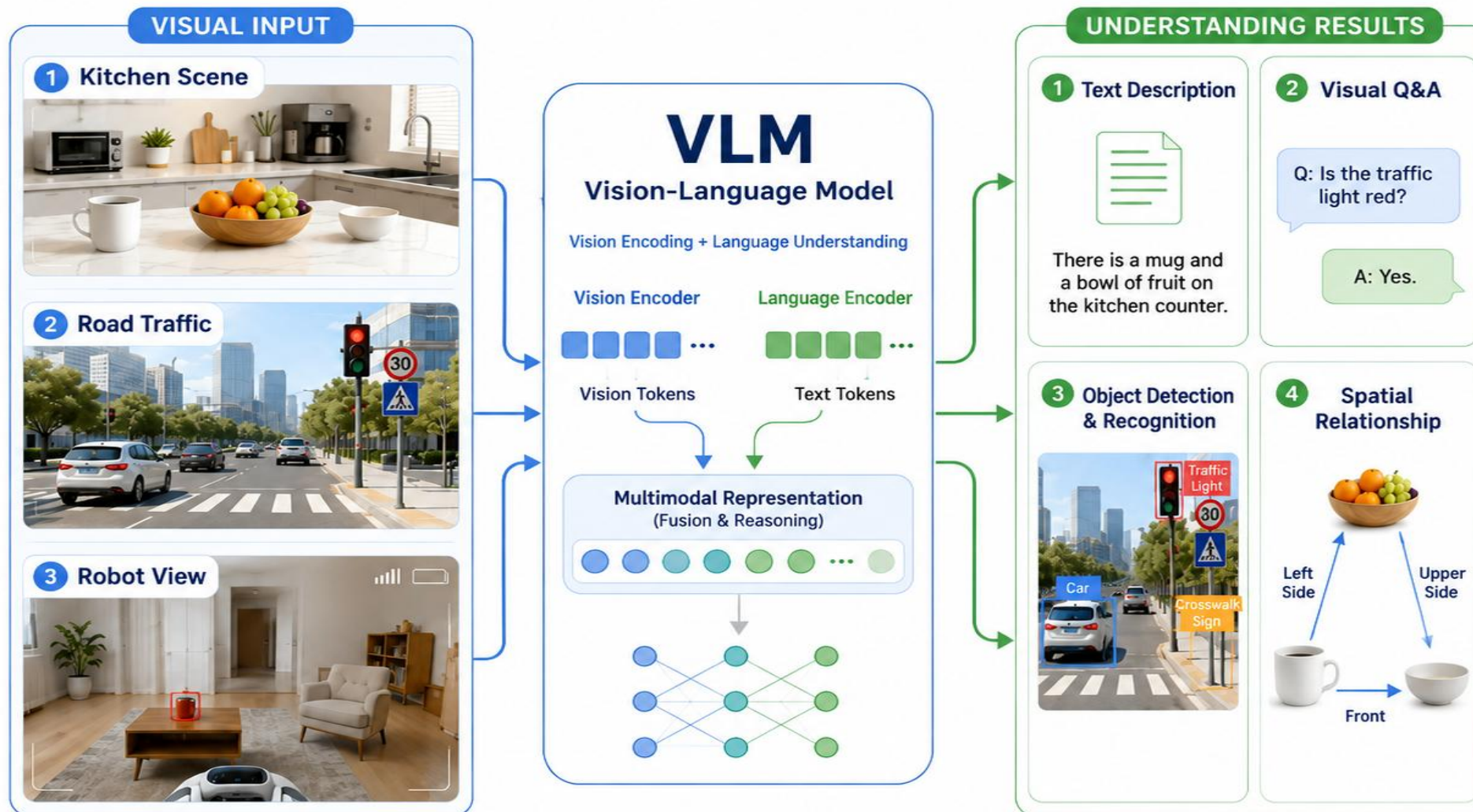
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- ✦ Key Pretraining:
 - ❑ Self-supervised Pretraining (single-stream and multi-stream) e.g., VL-BERT, ViLBERT
 - ❑ Contrastive Pretraining e.g., CLIP
- ✦ VLM-VLA Relationship:
 - ❑ VLM architectures transfer to VLAs
 - ❑ VLMs act as high-level planners

Some examples of VLM



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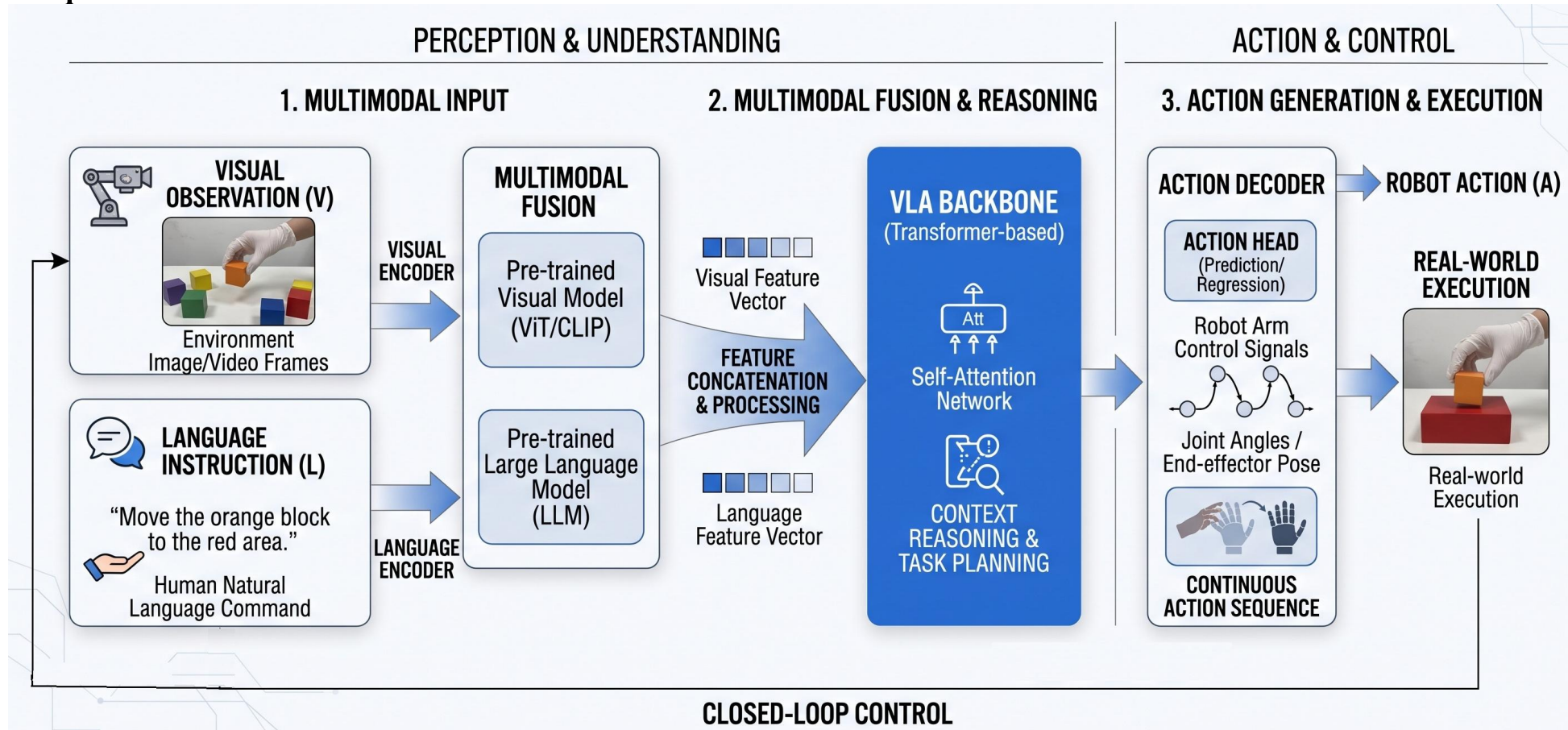


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What is VLA?

- ❑ Input: visual observations & natural language instructions & sensory modalities
- ❑ Output: control commands

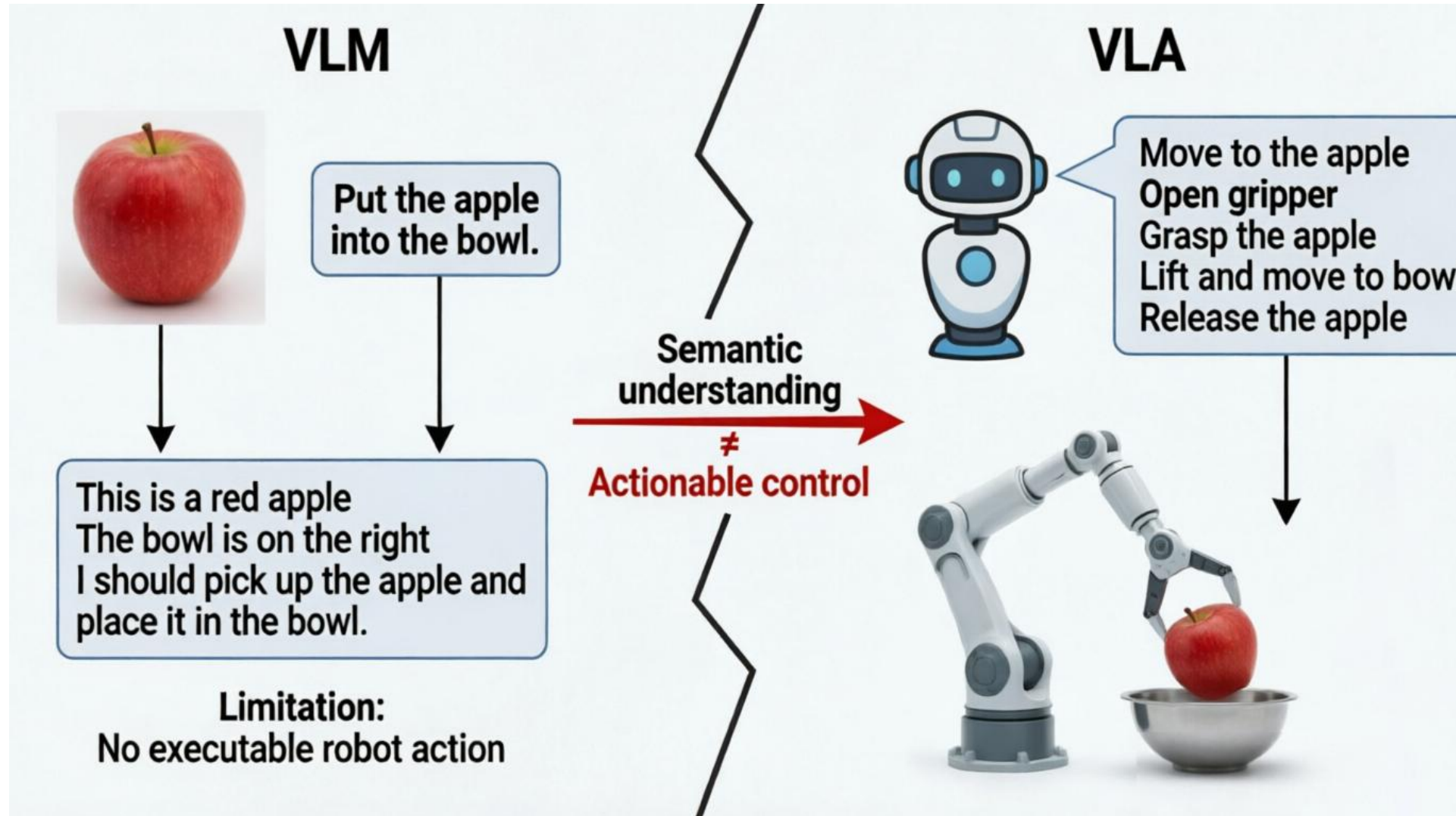


Why Not Directly Use VLMs?



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02

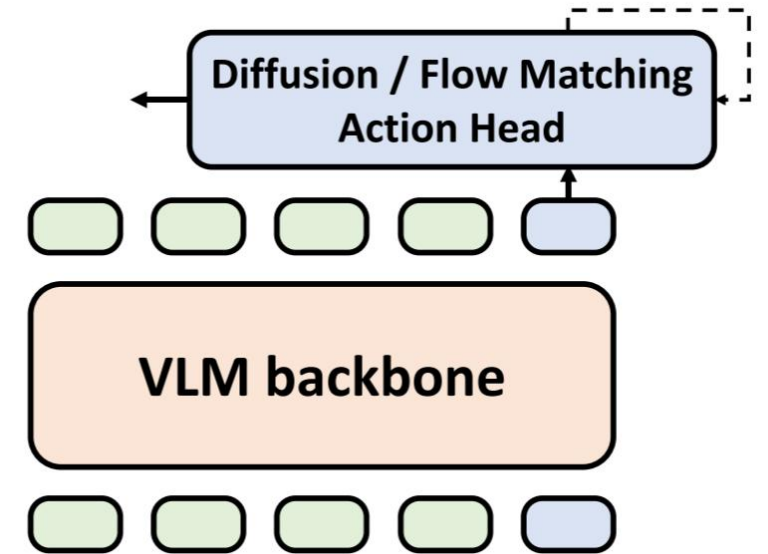
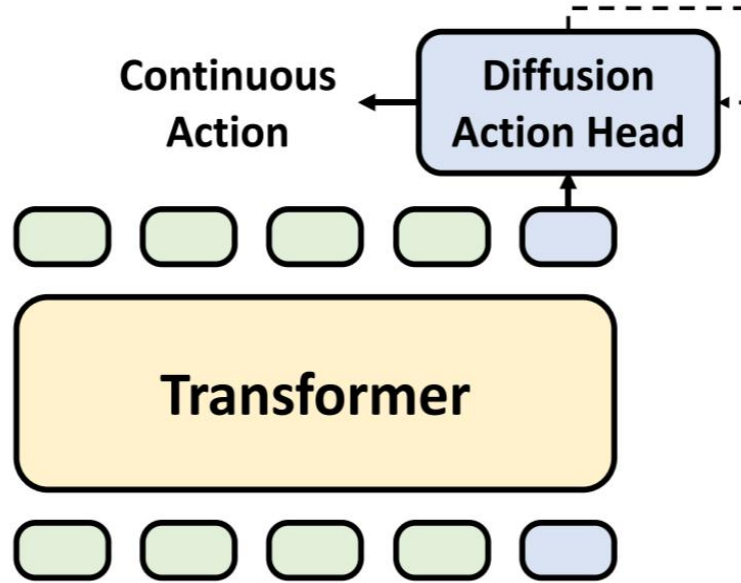
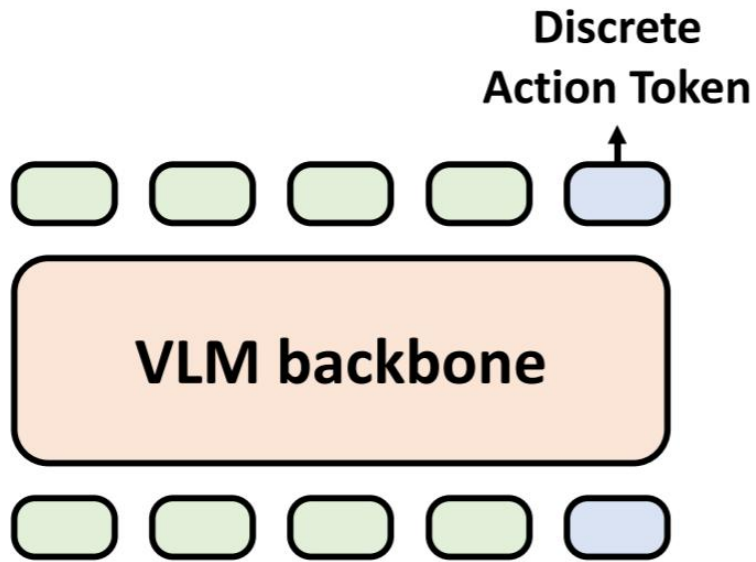
Dissect the Architectures

Three Core VLA Architectures



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(1) VLM + Discrete Action Token. (2) Transformer + Diffusion Action Head. (3) VLM + Flow Matching Action Head.

e.g., OpenVLA

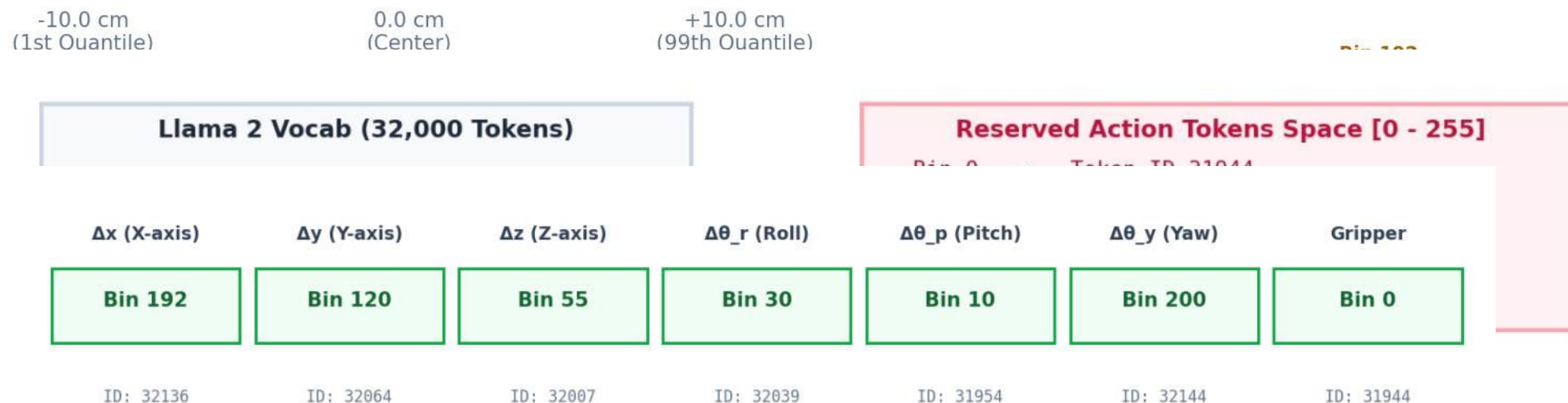
e.g., Octo

e.g., π_0

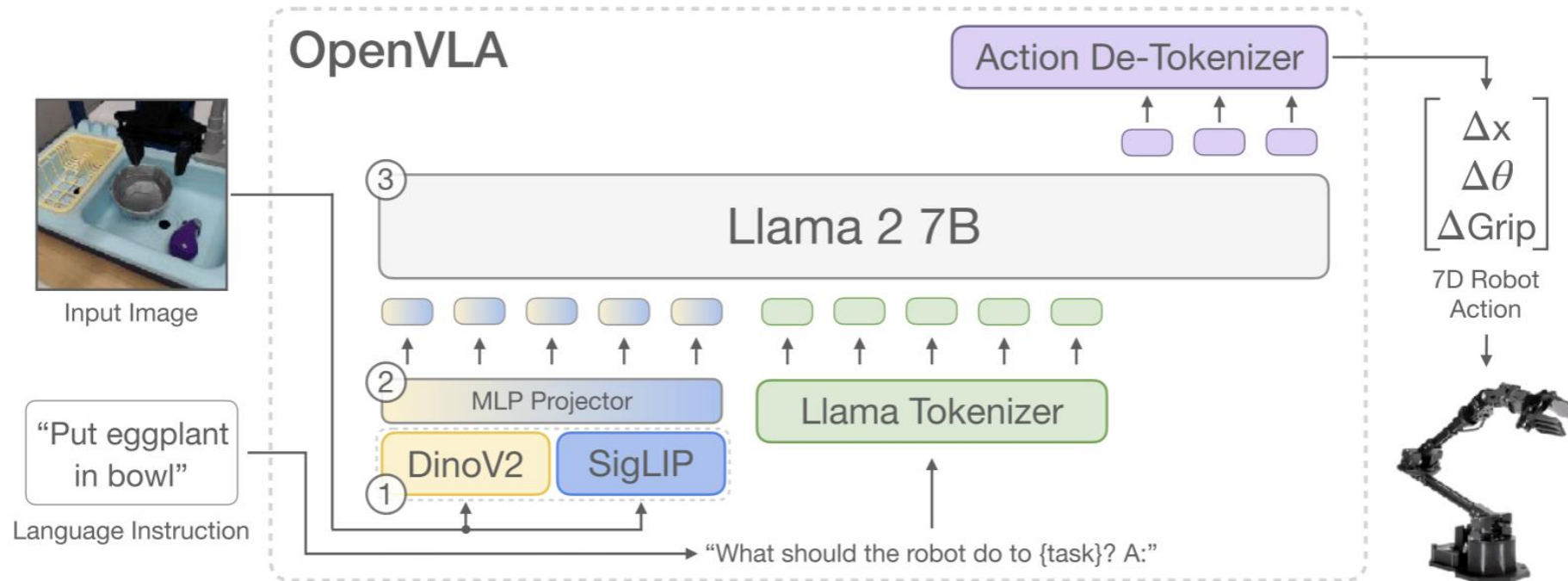
Preliminaries: discretized action tokens obtained via binning

- ❑ Discretize each dimension of the action space into bins (**typically 256**).
- ❑ Each bin ID is treated as a discrete token.
- ❑ Detailed Steps:

① Define Action Boundaries & Discretize into Bins

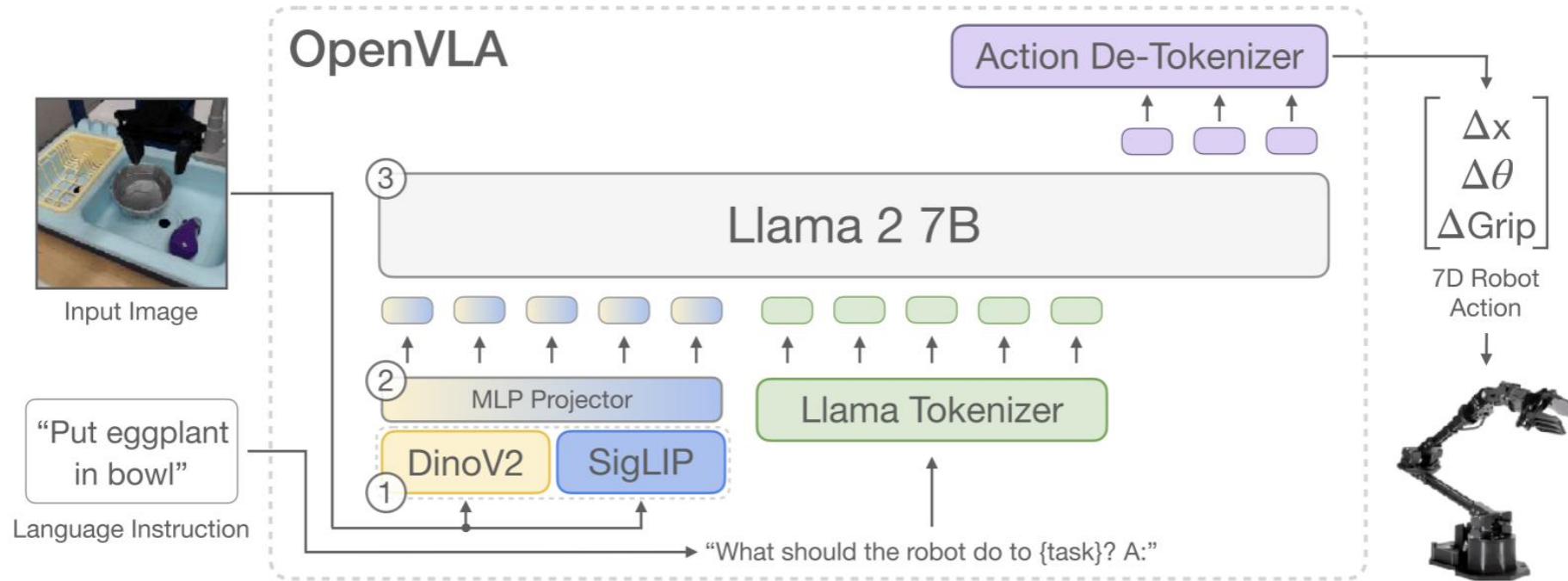


VLM + Discrete Action Token (OpenVLA)



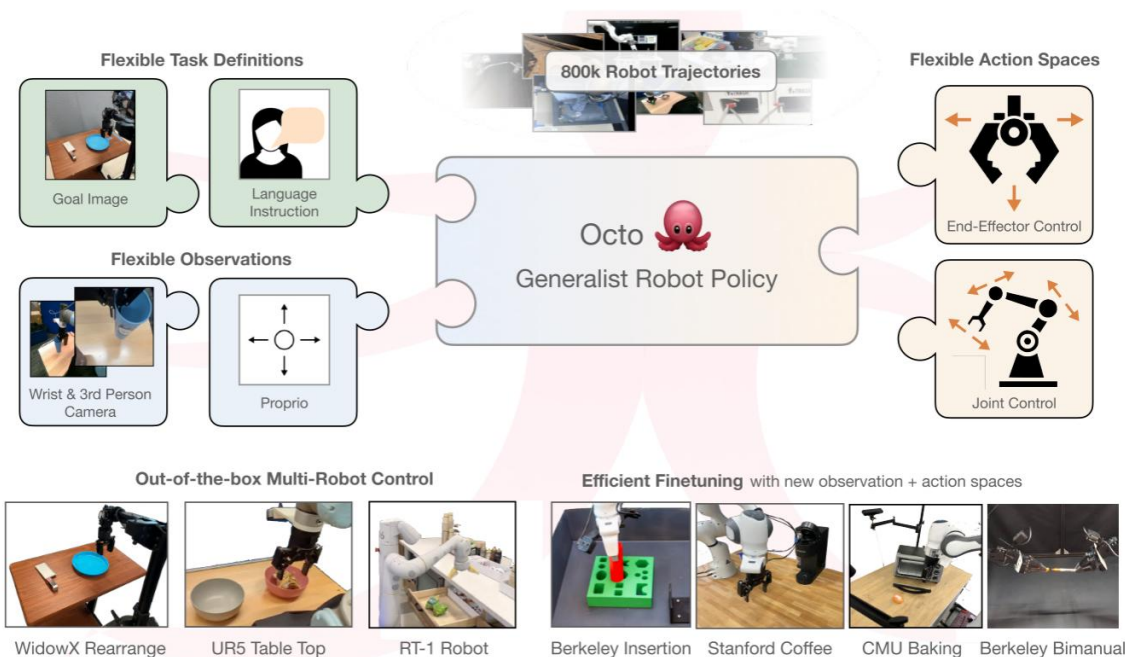
- ❑ Parameters: 7B
- ❑ Training Data: 970k real-world robot demonstrations (Open X-Embodiment)
- ❑ Full-parameter training: Unfreeze all parameters in OpenVLA
- ❑ Training Object & Loss: Next-token prediction with action-only cross-entropy loss

VLM + Discrete Action Token (OpenVLA)



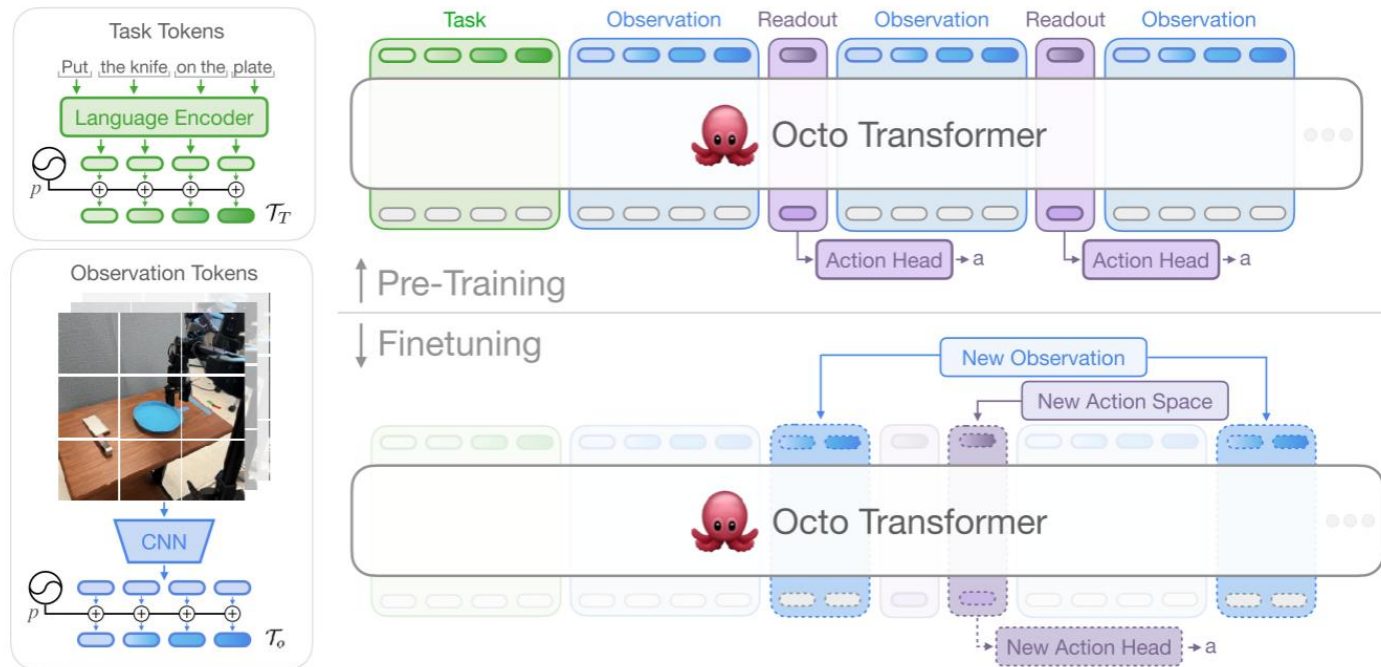
- ❑ Visual Encoder: 600M SigLIP (High-level semantic features) + Dino V2 (low-level spatial geometric details), Concatenated channel-wise
- ❑ Language Model Backbone: 7B Llama 2
- ❑ MLP projector: 2-layer
- ❑ Self-Attention Mechanism

Transformer + Diffusion Action Head (Octo)



- ❑ Training Data: 800K robot episodes (Open X-Embodiment)
- ❑ Flexible Observations
- ❑ Flexible Tasks
- ❑ Flexible Actions
- ❑ Parameters: Octo-Small (27M) / Octo-Base (93M)

Transformer + Diffusion Action Head (Octo)



□ Causal Attention & Block-wise Attention

□ Task Block: Bidirectional Attention

□ Observation Block: $\mathcal{T}_{o,t}$ only attends to $\mathcal{T}_{o,0:t}$ & $\mathcal{T}_T$

□ Read Tokens: $\mathcal{T}_{R,t}$ attends to $\mathcal{T}_o$ & $\mathcal{T}_T$ before it in the sequence (compact vector embedding of the observation sequence)

□ Text Encoder: t5-base (111M)

□ Visual Encoder: a lightweight CNN

□ Input Tokenizers: $\ell, g, (o_1, \dots, o_H) \rightarrow [\mathcal{T}_\ell, \mathcal{T}_g, \mathcal{T}_o]$

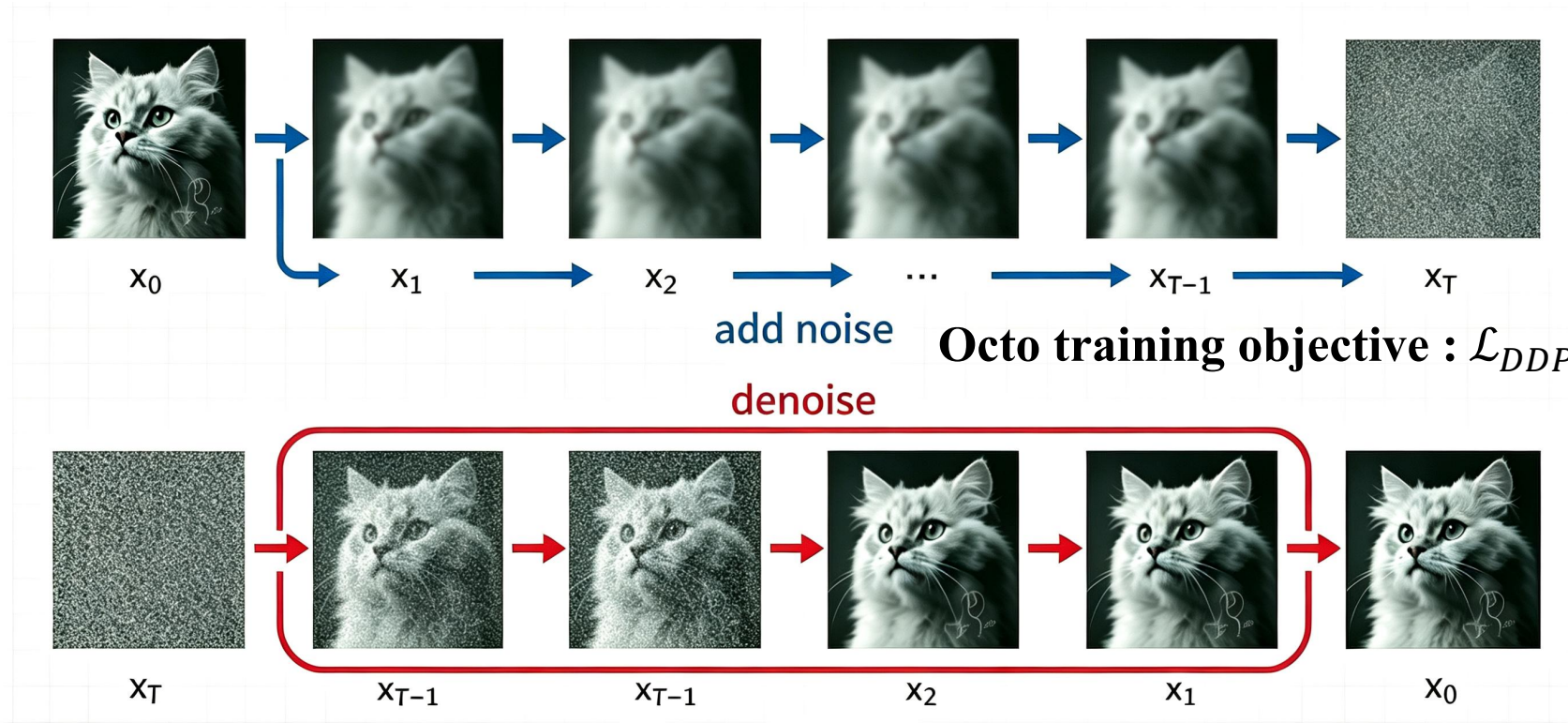
□ Learnable position embeddings p

□ Transformer Backbone:
 $e_\ell, e_g, e_o = T(\mathcal{T}_\ell, \mathcal{T}_g, \mathcal{T}_o)$

□ Readout Heads: $R(e) \rightarrow a$

Transformer + Diffusion Action Head (Octo)

✦ Training objective (DDPM)

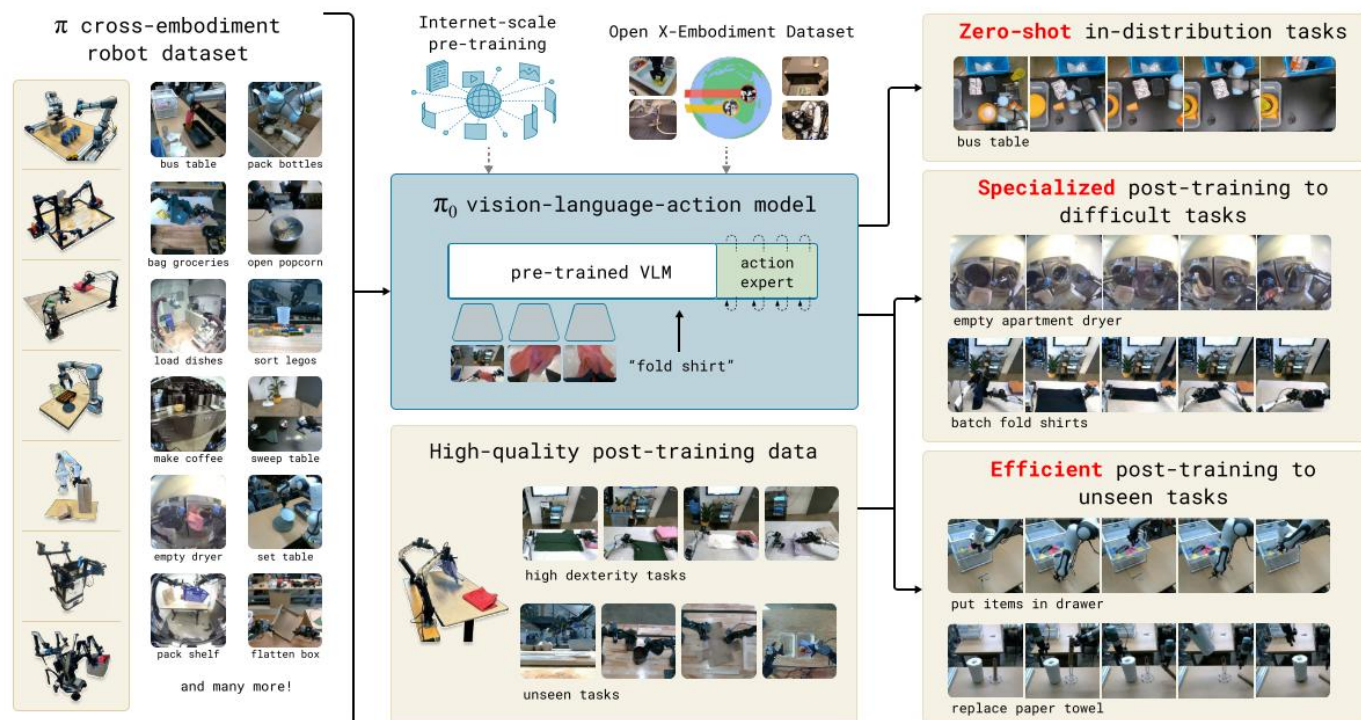


Octo training objective : $\mathcal{L}_{DDPM} = \mathbb{E}_{t,\epsilon,k} [\|\epsilon - \epsilon_\theta(a_t^k, e, k)\|^2]$

DDPM training objective: $L_{simple}(\theta) := \mathbb{E}_{t,x_0,\epsilon} [\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2]$

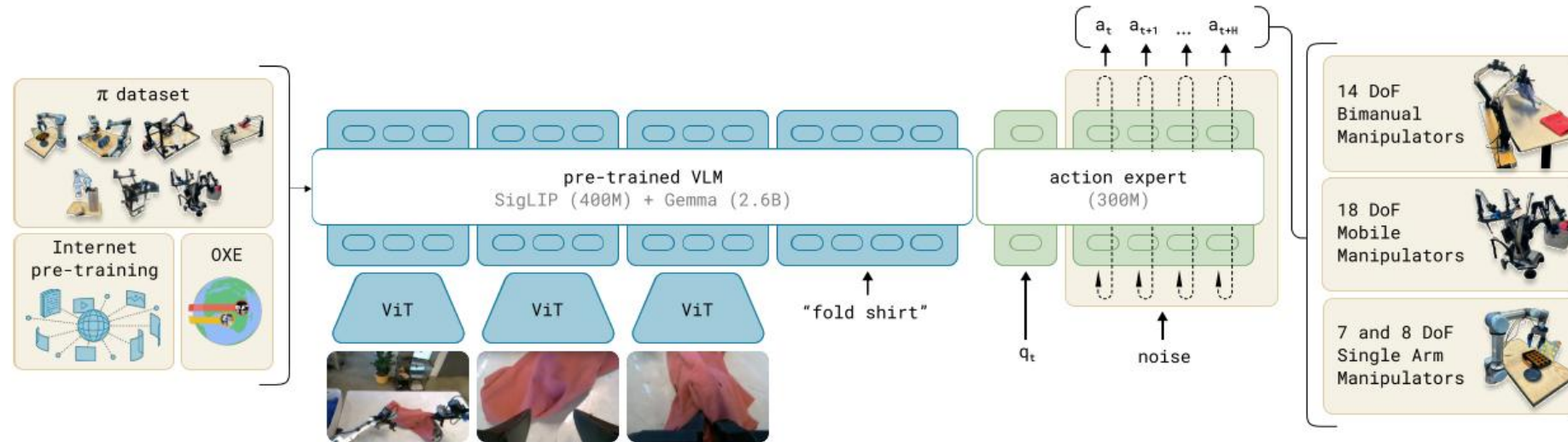
Denoising: $x^{k-1} = \alpha(x^k - \gamma \epsilon_\theta(x^k, e, k) + \mathcal{N}(0, \sigma^2 I)) \quad x^k \sim \mathcal{N}(0, I)$

VLM + Flow Matching Action Head (π_0)



- ❑ Parameters: 3.3B
- ❑ Training Data: 10,000 hours of dexterous manipulation data + open-source data
- ❑ Control Frequency: 50 Hz
- ❑ Training Paradigm: pre-training + post-training

VLM + Flow Matching Action Head (π_0)



❑ Base Model: 3B PaliGemma (SigCLP + Gemma) + 300M Action expert

❑ Input: $o_t = [I_t^1, \dots, I_t^n, \ell_t, q_t]$ $n = 2 \text{ or } 3$

❑ Output: $A_t = [a_t, a_{t+1}, \dots, a_{t+H-1}] \sim p(A_t | o_t)$ $H = 50$

❑ $[I_t^1, \dots, I_t^n, \ell_t] \rightarrow \text{VLM}$ $[q_t, A_t^\tau] \rightarrow \text{Action expert}$

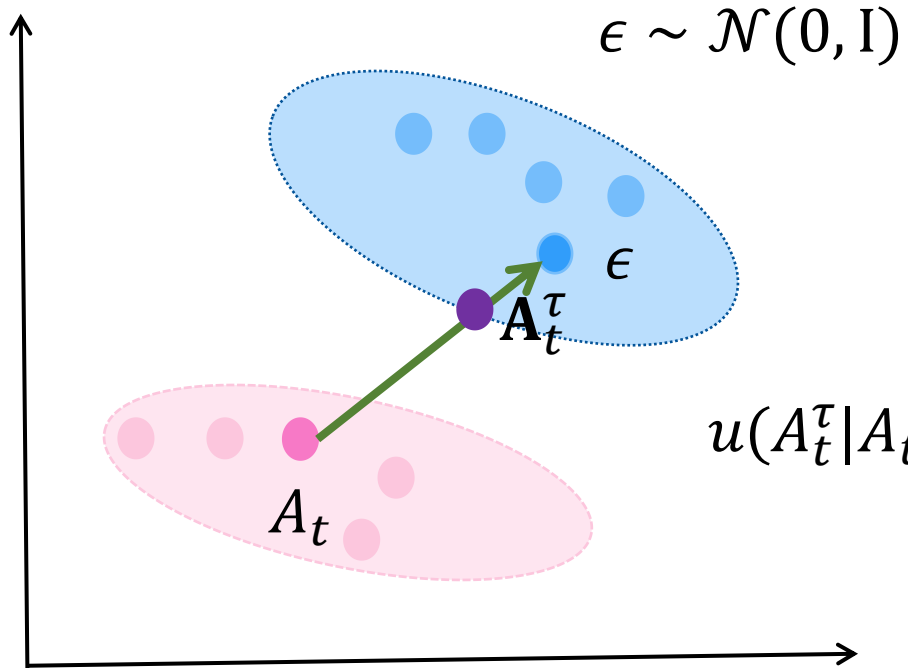


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VLM + Flow Matching Action Head (π_0)

✦ Training objective (Flow Matching)



$$q(A_t^\tau | A_t) = \mathcal{N}(\tau A_t, (1 - \tau)I)$$

$$A_t^\tau = \tau A_t + (1 - \tau)\epsilon \quad \tau \in [0, 1]$$

$$u(A_t^\tau | A_t) = \epsilon - A_t \quad A_t = [a_t, a_{t+1}, \dots, a_{t+H-1}]$$

$$L^\tau(\theta) = \mathbb{E}_{p(A_t | o_t), q(A_t^\tau | A_t)} ||v_\theta(A_t^\tau, o_t) - u(A_t^\tau | A_t)||^2$$

$$\textit{Inference} : A_t^{\tau+\delta} = A_t^\tau + \delta v_\theta(A_t^\tau, o_t)$$

$$A_t^0 \sim \mathcal{N}(0, I)$$

$$\tau = 0 \rightarrow \tau = 1$$

03

Understand the Training Data

Datasets for VLA



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1. HUMAN DATASETS

Ego4D

EPIC-KITCHENS

ARCTIC

OAKINK2

EMBODIMENT GAP

2. SIMULATION DATASETS

RoboTurk

MimicGen

DexMimicGen

SIM-TO-REAL GAP

3. REAL ROBOT DATASETS

QT-Opt

BridgeData V2

DROID

Open X-Embodiment (OXE)

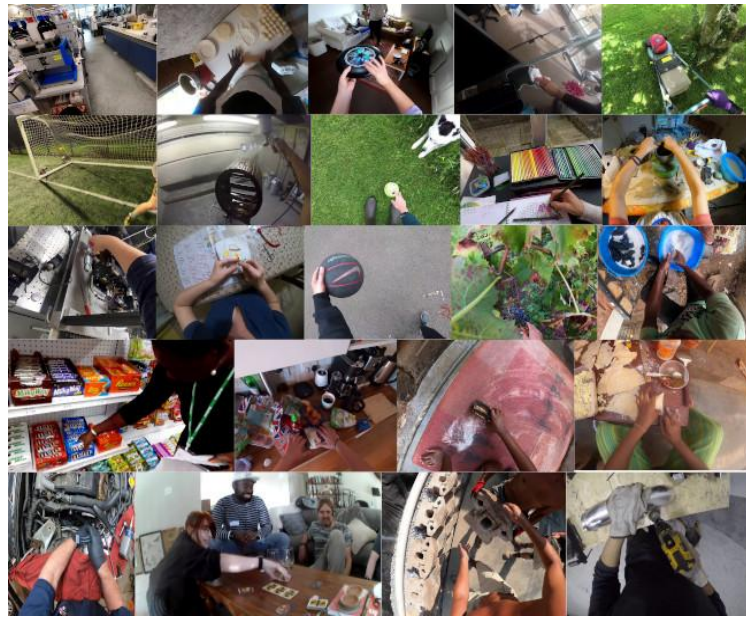
SCARCITY & HETEROGENEITY

Datasets for VLA- Some examples

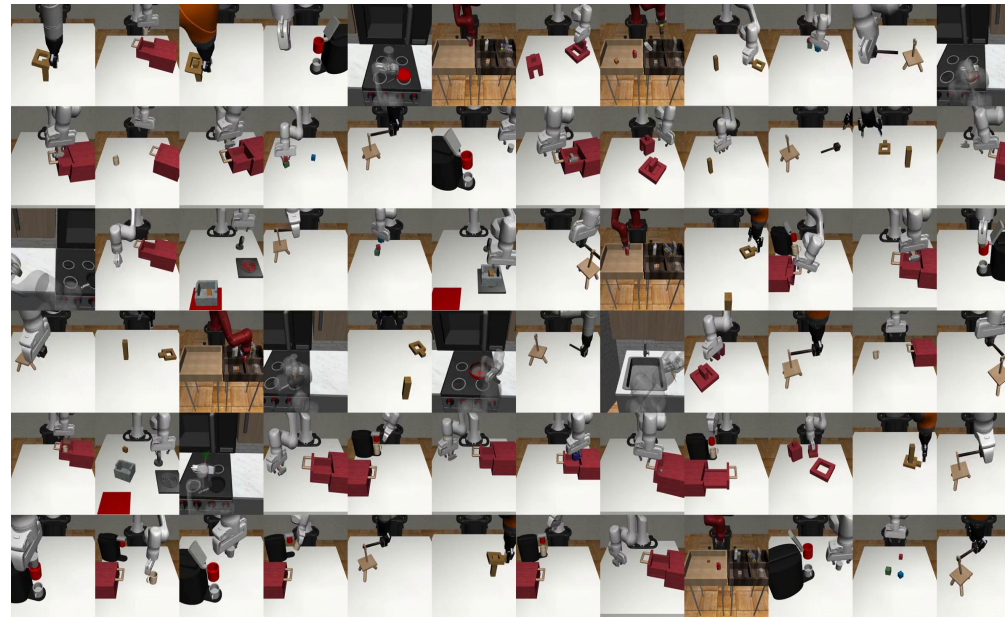


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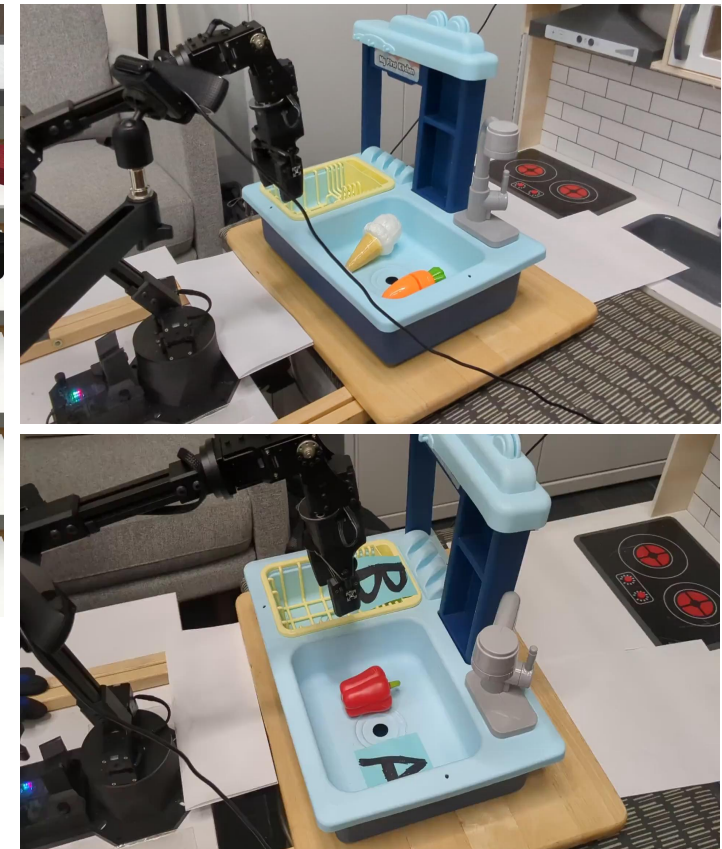
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❑ Human datasets (Ego4D)



❑ Simulation datasets (MimicGen)



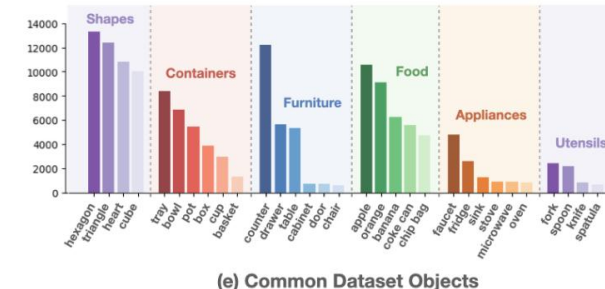
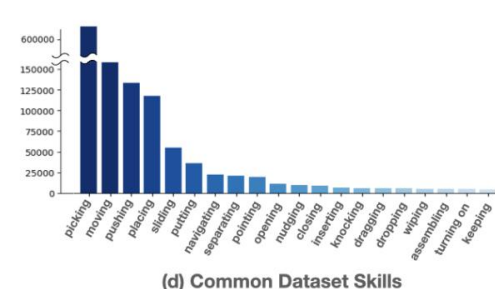
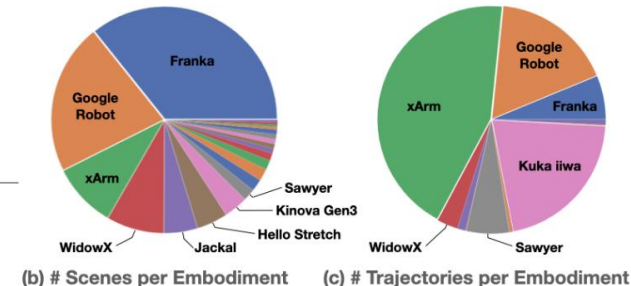
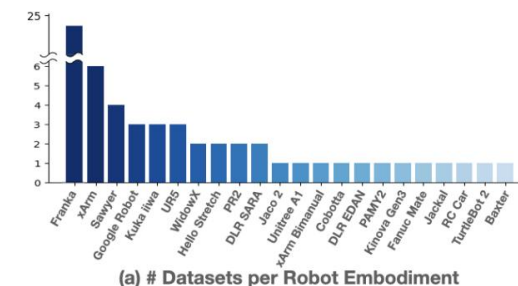
❑ Real robot datasets (OXE)

Open X-Embodiment: The ImageNet Moment for VLA



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- ❑ 22 robot embodiments
- ❑ 1M+ real robot trajectories

- ❑ 527 skills (160266 tasks)
- ❑ 34 robotic research labs

04

Understand the Training Methods

Two-Stage Pipeline



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□ Pre-training

- Acquire **general capabilities** and promote **transferability across diverse robotic embodiments**.
- Robot demonstrations + Web-scale vision-language corpora + Structured annotations
→ Improve semantic understanding and visuomotor grounding
- Initialized from VLM backbones

□ Post-training

- Refined using **high-quality robot demonstration data** to improve performance on **specific downstream tasks**.
- Requires high-quality, robot and task-specific data
- Full Fine-Tuning vs. LoRA Fine-Tuning vs. Action-Head-Only Fine-Tuning

References



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- ✓ Ma, Yuen, et al. "A survey on vision–language–action models for embodied ai." IEEE Transactions on Neural Networks and Learning Systems (2026).
- ✓ Kawaharazuka, Kento, et al. "Vision-language-action models for robotics: A review towards real-world applications." IEEE Access (2025).



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THANKS